

The Role of Explainable AI (XAI) In Enhancing Transparency and Trust in NLP-Powered Educational Systems

Priya Kumawat¹, Dr Pradeep Singh Shaktawat²

¹Assistant Professor, Department of Computer Science, Aishwariya Post Graduate College, Udaipur, Rajasthan, India

²Assistant Professor, Department of Computer Science and IT, JRN Rajasthan Vidyapeeth Deemed to be University, Rajasthan, India

ARTICLE INFO

Article History:

Accepted : 07 July 2025

Published: 22 August 2025

Publication Issue

Volume 11, Issue 4

July-August-2025

Page Number

432-438

ABSTRACT

As Artificial Intelligence (AI) systems become increasingly integrated into educational settings, the demand for transparency and trustworthiness has grown. Natural Language Processing (NLP)-powered applications such as intelligent tutoring systems, automated essay scoring, and educational chatbots offer significant benefits for personalized learning, yet often operate as “black boxes.” The lack of explainability in these models can undermine user trust, raise ethical concerns, and limit their effective use in classrooms. Explainable Artificial Intelligence (XAI) offers a critical solution by making AI decisions interpretable and justifiable to end-users. This review explores the role of XAI in enhancing transparency and trust within NLP-powered educational systems. It examines core challenges faced by educators and learners when using opaque AI, including bias, accountability, and adoption resistance. The paper reviews XAI techniques such as feature attribution, attention visualization, and open learner models that provide insights into model behavior. Real-world applications like the iRead literacy tutor and AI chatbots for feedback analysis illustrate how XAI can improve stakeholder confidence and system usability. The paper also outlines future research directions, emphasizing the need for user-centered explanations, multimodal transparency, and standardized evaluation frameworks. Ultimately, the integration of XAI into educational NLP tools is not merely a technical enhancement—it is essential for building ethical, effective, and human-aligned AI systems in education. By making AI outputs understandable and actionable, XAI bridges the gap between powerful algorithms and pedagogical trust.

Keywords: Explainable AI (XAI), Natural Language Processing (NLP), Educational Technology, Intelligent Tutoring Systems, Trust in AI

Introduction

Artificial intelligence (AI) – especially Natural Language Processing (NLP) – is increasingly embedded in educational software, from intelligent tutoring systems and automated grading to conversational chatbots. These AI-driven systems promise personalized learning at scale, but they often operate as “black boxes,” making decisions without revealing their reasoning. This lack of transparency can undermine trust among educators and students. Explainable AI (XAI) in education refers to AI systems that make their decision-making processes visible and understandable. By exposing how an NLP model arrived at a recommendation or score, XAI helps stakeholders understand the AI’s reasoning, supporting fairer and more effective learning experiences. In high-stakes environments like education, such transparency is critical for using AI tools responsibly and ensuring user acceptance.

Building trust in AI-powered educational systems is not just a technical issue but also a pedagogical and ethical imperative. Teachers need confidence that an automated system’s suggestions or grades are grounded in valid reasoning, and students need to feel they are treated fairly and can learn from AI feedback. Reports of large language models producing biased or incorrect outputs have heightened awareness of these trust issues. Consequently, there is growing recognition that we must “open the black box” of educational AI. This review examines how XAI can enhance transparency and trust in NLP-powered educational systems. It synthesizes current literature, discusses challenges when AI lacks explainability, surveys emerging XAI frameworks, highlights illustrative applications, and outlines future research directions to guide the development of transparent, trustworthy AI in education.

Literature Review

AI in Education and NLP Applications: In the past decade, numerous studies have demonstrated AI’s potential to transform education. Torfi et al. (2020)

note that advances in deep learning have greatly improved NLP’s ability to interpret text, enabling more sophisticated educational applications. Such progress has paved the way for tools like automated essay scoring, dialogue-based tutoring, and sentiment analysis of student feedback. Huang et al. (2023) found a sharp rise in research on AI for language education – covering topics from intelligent tutoring systems and automated writing evaluation to speech recognition for pronunciation training. These applications illustrate AI’s growing versatility: from assessing free-text answers in science classes to guiding language learners, AI is increasingly intertwined with educational practice.

Benefits and Emerging Tools: The integration of large language models has further accelerated interest in AI for education. Powerful generative models like OpenAI’s GPT series can produce human-like responses and support a range of educational activities. Hariri (2023) notes that ChatGPT and similar models have transformative potential in domains from content creation to personalized tutoring. One study by Li and Xing (2021) found that a GPT-2 based discussion agent significantly enhanced engagement in a massive open online course (MOOC) forum by providing timely, contextually relevant replies to learner posts. These examples suggest that AI, when thoughtfully applied, can augment human educators and provide more personalized learning experiences at scale.

Challenges Highlighted in Prior Work: Despite these advances, researchers caution about challenges that directly motivate the need for XAI. Many NLP models used in education are complex and opaque, making it hard to diagnose errors or biases. Hariri (2023) notes that even advanced models like ChatGPT can exhibit biased or inappropriate responses, raising ethical concerns about their unregulated use. Likewise, Shaik et al. (2022) review NLP-based feedback analysis and highlight technical hurdles such as ambiguous language (e.g. sarcasm) and the need for large domain-specific datasets to improve accuracy. On the other

hand, AI's capabilities are evident – for instance, an ML model in one study closely matched human grading of science responses – yet teachers may hesitate to trust such automated judgments without clear justification. In summary, while AI's role in education is rapidly expanding, the lack of transparency in current systems presents a barrier. Danilevsky et al. (2020) stress that improving model interpretability is essential for users to trust and effectively utilize NLP models. Some scholars argue that research so far has not adequately addressed issues of fairness, accountability, and transparency in AI-driven education, calling for a stronger focus on these aspects.

Challenges in AI-Driven Education Without XAI

Implementing AI in educational settings without sufficient explainability can lead to numerous challenges. Key concerns identified by educators and researchers include:

- **Lack of Transparency Undermining Trust:** When an AI system's decision process is opaque, teachers and students may doubt its outputs. For example, if an automated essay grader returns a low score without explanation, the student is left guessing which parts of the essay were subpar. This opacity makes it hard for learners to improve and for instructors to validate the system's correctness. Trust in the technology erodes if users cannot understand how or why a decision was made.
- **Bias and Fairness Issues:** Black-box NLP models can inadvertently perpetuate biases present in their training data, leading to unfair outcomes. For instance, a tutoring system might misinterpret the dialect or slang of a particular student group, resulting in misleading feedback. Without XAI, such biases might remain hidden. Hariri (2023) observes that large language models have shown biased behaviors at times, and without transparency, correcting these issues is extremely difficult. Unexplained biases could

undermine the equity and inclusivity that education strives for.

- **Accountability and Ethical Concerns:** Educators are concerned about who is accountable when an AI system makes a flawed recommendation. If a learning analytics tool flags a student as “at risk” based on a proprietary algorithm, on what grounds can a counselor trust or contest that decision? Lack of explainability complicates compliance with emerging AI ethics guidelines and regulations. Notably, regulations like the EU AI Act will mandate a level of transparency for high-risk AI applications, which would include many educational tools. Without XAI, schools risk deploying AI that fails to meet such accountability standards, leading to ethical or legal issues.
- **Resistance from Educators and Students:** Teachers may resist integrating AI systems that they do not understand. A seasoned educator has intuition about their students that an opaque algorithm may not capture; if the AI's actions cannot be explained, the teacher is likely to mistrust or ignore its recommendations. Likewise, students might disengage or react negatively to AI-provided feedback that seems arbitrary. Studies suggest some users are initially skeptical of AI recommendations, but providing clear explanations significantly increases their willingness to trust and use the system.

In summary, without explainability, AI-driven educational systems risk being mistrusted and misused. They may produce results that are accurate on paper yet unusable in practice because stakeholders cannot validate or comprehend them. These challenges motivate the integration of XAI techniques to make AI decisions more transparent, accountable, and aligned with educational values.

XAI Solutions and Frameworks in Educational NLP

To address the above challenges, developers are incorporating explainability techniques into educational AI systems. XAI solutions in NLP aim to

make a model's logic visible – by attributing importance to input features, simplifying model behavior, or providing human-readable justifications. Several noteworthy approaches and frameworks include:

- **Feature Attribution and Visualization:** Tools like LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (Shapley Additive Explanations) highlight which parts of an input most influenced a model's output, offering insight into why the AI responded a certain way. In an educational context, this could mean highlighting the phrases in an essay that led an automated grader to deduct points. Similarly, visualizing model internals (for example, displaying attention weights in a transformer-based language model) can show what tokens or concepts the AI focused on. An instructor using an AI tutor might see that the system placed heavy “attention” on a specific step of a student's solution, explaining why it gave a particular hint. Interactive interfaces like Google's What-If Tool allow users to tweak inputs and observe changes in predictions, further demystifying the model's behavior.
- **Example-Based Explanations:** Another XAI approach is to provide human-understandable examples or counterfactual cases. For instance, an AI writing tutor could retrieve a high-scoring example essay to compare with the student's essay, highlighting differences in structure or content. This contextualizes the feedback by showing “what good looks like.” Some toolkits (e.g. IBM's AI Explainability 360) facilitate such contrastive explanations, helping translate model decisions into concrete suggestions (“Your answer missed key point X, unlike this example answer”). By referencing real or hypothetical examples, the AI's reasoning becomes more transparent and relatable to the learner.
- **Open Learner Models:** Many intelligent tutoring systems use Open Learner Models (OLMs) as a

built-in explainability feature. An OLM displays the system's assessment of a student's mastery – for example, a dashboard showing which topics a student has mastered or where they are struggling. Exposing the AI's internal student model allows learners and teachers to see *why* the system is giving certain recommendations. They can verify that, say, the tutor keeps focusing on fractions because the student's mastery level for “fractions” remains low. By making student models visible (and even editable in some cases), OLMs foster transparency and trust: learners can correct the system's misunderstandings, and teachers can better align their instruction with the AI's insights.

Developers often address the accuracy–interpretability trade-off by using complex models for predictions but wrapping them with an explanation layer (e.g. an interpretable proxy or a natural-language justification for each output). In practice, making the AI's reasoning clearer allows teachers to validate or override AI outputs and helps students learn from AI feedback. Ultimately, transparency turns the AI system into a more trustworthy assistant rather than an inscrutable oracle.

Case Studies and Applications

To illustrate how XAI can enhance transparency and trust, we consider two examples where explainability was applied to NLP-powered educational systems:

Case 1: The iRead Project (Intelligent Tutoring for Literacy). A compelling example is the *iRead* project – an EU-funded intelligent tutor designed to support children's reading skills. The developers of iRead placed a strong emphasis on explainability from the outset. The system adapts to individual learners by assessing their performance on literacy tasks, and it makes these adaptations transparent to users. If a student struggles with a particular skill, iRead adjusts upcoming exercises and explicitly communicates that it's focusing on that weak area. Across multiple pilot programs, this transparent design yielded significant

benefits. Karpouzis (2024) reports that iRead's explainable feedback improved students' motivation and engagement, and the clarity of its adaptive interventions eased teachers' concerns about using the system in class. When teachers could see why the AI chose a certain exercise for a student – for example, “the tutor is giving extra practice on inferencing because the student had difficulty with that skill” – they felt more confident integrating the tool into their instruction. The iRead case demonstrates that building explainability into an educational NLP system can lead to more effective learning and greater stakeholder trust, as the AI's actions align with human expectations and pedagogical reasoning.

Case 2: AI Chatbot for Student Feedback (Hubert.ai).

Another example involves using an AI chatbot to collect and analyze student feedback. In one university trial, a chatbot called *Hubert.ai* was deployed to gather students' comments about a course. The NLP-driven bot could quickly summarize common themes in hundreds of open-ended responses, providing the instructor with an overview of what students found positive or challenging. This automation saved time and helped identify issues faster. In fact, researchers have observed that successful educational chatbots typically include features to make their NLP analysis transparent to instructors (for example, dashboards highlighting prevalent keywords in student comments). However, without transparency into how the bot categorized and interpreted the comments, some instructors were unsure how much to trust its summaries. In response, developers added explainability features: the system began showing sample student comments for each detected theme and highlighting keywords it considered important. This allowed instructors to verify that, for example, “many students mentioned *lecture pace* as an issue” was backed by actual quotes. Once its inner workings were exposed, the instructors' confidence in the AI analysis increased. This case highlights that even when AI provides useful automation, explaining its workings is vital. Users are

more likely to trust and act on a chatbot's analysis if they can see the evidence and reasoning behind it, reinforcing the importance of XAI in such educational applications.

Future Research Directions

Moving forward, several research directions can further enhance transparency and trust in AI-driven education:

- **Bias Mitigation and Fairness:** Ensuring AI systems treat all students fairly is paramount. Future research should develop methods to detect and reduce biases in educational NLP models. XAI can assist by revealing where a model's predictions might be unfair or skewed. For example, explainability tools could flag if an automated grader consistently scores one group's submissions lower. Recent work supports this – a review of XAI for student interventions found that transparent models improved identification of at-risk learners and personalized support. Research is needed on techniques like bias audits for educational datasets, debiasing algorithms, and fairness-focused interventions so that AI recommendations do not inadvertently disadvantage any group of learners.
- **Human-Centered Explanations and AI Literacy:** Future work should focus on designing explanations that stakeholders such as teachers and students find intuitive and useful. This involves user-centered studies to determine what kinds of explanations (visual dashboards, simplified language justifications, etc.) best improve understanding and trust. Efforts to build AI literacy go hand-in-hand with this – educators and learners need some familiarity with AI concepts to interpret AI outputs critically. Co-designing XAI tools with input from teachers and students, and conducting longitudinal studies on how they use these tools over time, will help ensure that explainability features truly meet users' needs.

- **Standards and Policy for XAI:** The field would benefit from agreed-upon standards for explainability in educational AI. Researchers and policymakers could define metrics or an “explainability score” to rate a system’s transparency. Such standards would help schools choose AI tools that meet transparency requirements. Policy frameworks (such as the EU AI Act) are already pushing for greater transparency; research can support these efforts by determining what information about an AI’s logic should be disclosed in educational contexts to satisfy accountability and ethical guidelines. Additionally, developing evaluation benchmarks to test whether explanations actually improve user trust and learning outcomes is an important aspect of this direction.
- **Explainability for Complex AI (LLMs):** As large language models become prevalent in education (e.g. for essay feedback or tutoring via chatbots), new XAI techniques are needed to interpret these complex models. Today’s most advanced NLP models are very accurate but often operate as black boxes. Research might explore creating simpler surrogate models or using probing methods to extract understandable patterns from an LLM’s hidden layers. The goal is to enable even advanced AI systems to articulate reasons for their outputs in educational terms. Ensuring that future AI tutors and assistants can explain *why* they made a certain suggestion or evaluation will be crucial as their usage grows in classrooms.

Conclusion

AI-powered educational systems, particularly those leveraging NLP, hold great promise for enriching learning experiences and improving efficiency. However, to realize this promise, these systems must be trusted by the people who use them. Explainable AI is a critical enabler of that trust, as it addresses the opacity that hinders AI adoption in classrooms. Without explainability, user distrust, biased outcomes,

and ethical dilemmas can arise; conversely, integrating explainability techniques makes AI a more reliable partner in education – one whose suggestions can be understood and vetted by teachers and students.

The case studies and research discussed in this paper illustrate that transparency and interpretability should be treated as core features of educational AI systems, not afterthoughts. When students understand why a tutoring system gave them a certain hint, or when teachers see the evidence behind an AI’s assessment, the technology becomes more credible and effective. The iRead project showed improved learning outcomes and teacher acceptance with explainability built in, and similar findings are emerging across other applications. Open questions remain on how best to design explanations for different users and how to explain the most complex models, but progress is clearly moving in this direction.

Explainable AI plays a pivotal role in ensuring that NLP-powered educational systems are not only intelligent but also transparent and trustworthy. By making AI decisions visible and understandable, we empower educators to confidently integrate AI into their teaching and enable students to learn with the aid of AI in a meaningful way. Continued collaboration between AI developers, educators, and policymakers on explainability will be essential to harness the benefits of AI in education while upholding fairness, accountability, and trust.

References

- [1]. Aleedy, M., Atwell, E., & Meshoul, S. (2022). Using AI chatbots in education: Recent advances, challenges and use case. In *Artificial Intelligence and Sustainable Computing (Proceedings of ICSISCET 2021)* (pp. 661–675). Springer, Singapore.
- [2]. Danilevsky, M., Qian, K., Aharonov, R., Katsis, Y., Kawas, B., & Sen, P. (2020). A survey of the

- state of explainable AI for natural language processing. arXiv preprint, arXiv:2010.00711.
- [3]. European Commission. (2024, November 7). Insights from the community workshop on explainable AI in education. European Education Area – News. Retrieved from <https://education.ec.europa.eu/news/insights-from-the-community-workshop-on-explainable-ai-in-education>
- [4]. Hariri, W. (2023). Unlocking the potential of ChatGPT: A comprehensive exploration of its applications, advantages, limitations, and future directions in natural language processing. arXiv preprint, arXiv:2304.02017.
- [5]. Huang, X., Zou, D., Cheng, G., Chen, X., & Xie, H. (2023). Trends, research issues and applications of artificial intelligence in language education. *Educational Technology & Society*, 26(1), 112–131.
- [6]. Karpouzis, K. (2024). Explainable AI for intelligent tutoring systems. In *Frontiers of Artificial Intelligence, Ethics, and Multidisciplinary Applications* (pp. 59–70). Springer, Cham.
- [7]. Li, C., & Xing, W. (2021). Natural language generation using deep learning to support MOOC learners. *International Journal of Artificial Intelligence in Education*, 31(1), 186–214.
- [8]. Liu, B., Li, C., Xu, X., Wang, J., Zheng, C., & Lu, Y. (2025). Using explainable AI (XAI) to identify and intervene with students in need: A review. In *Proceedings of the 2024 International Conference on Artificial Intelligence in Education (ICAIE)* (pp. 636–641). ACM.
- [9]. McKinsey & Company. (2024, November 26). Building AI trust: The key role of explainability. Retrieved from <https://www.mckinsey.com/capabilities/quantumblack/our-insights/building-ai-trust-the-key-role-of-explainability>
- [10]. Shaik, T., Tao, X., Li, Y., Dann, C., McDonald, J., Redmond, P., & Galligan, L. (2022). A review of the trends and challenges in adopting natural language processing methods for education feedback analysis. *IEEE Access*, 10, 56720–56739.
- [11]. Torfi, A., Shirvani, R. A., Keneshloo, Y., Tavaf, N., & Fox, E. A. (2020). Natural language processing advancements by deep learning: A survey. arXiv preprint, arXiv:2003.01200.